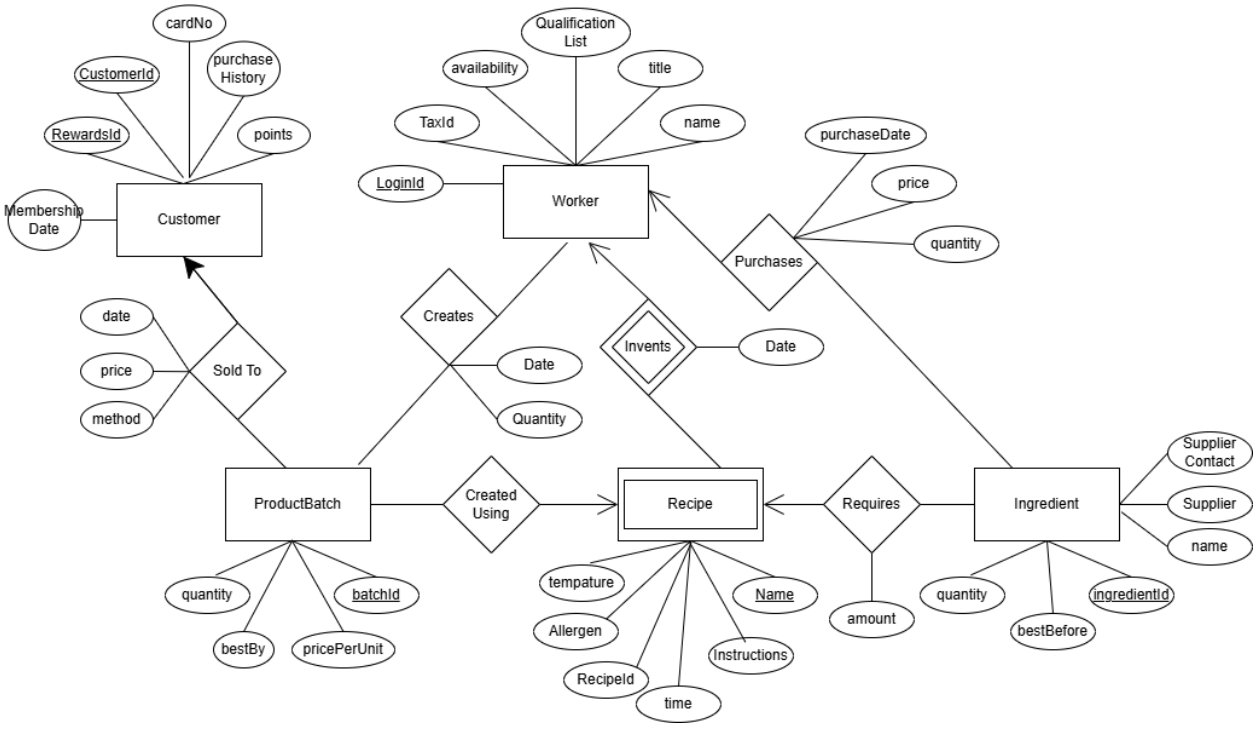
SmartBaker System

By: Jack Berry, Jonathon Houston, and Matt Settle-Rakay

Description

The following DDL statements represent our modified ER schema from Phase 1. We have omitted the weak entity set for rewards members and instead gave each customer an individual amount of points and a rewards ID. Following these CREATE TABLE statements, we have added additional DDL statements to insert sample data into each table.

Updated ER Schema



Relational Model

Customer(customerId, cardNo, purchaseHistory)
RewardsMember(rewardsId, points)
Joins(customerId, rewardsId, membershipDate)
Worker(loginId, taxId, availability, QualificationList, title, name)
ProductBatch(batchId, quantity, bestBy, pricePerUnit)

Recipe(recipeId, temperature, Allergen, time, Instructions, Name)
Ingredient(ingredientId, quantity, bestBefore, name, Supplier, SupplierContact)
SoldTo(customerId, batchId, date, price, method)
Creates(loginId, batchId)
CreatedUsing(batchId, recipeId, Date, Quantity)
Invents(loginId, recipeId, Date)
Requires(recipeId, ingredientId, amount)
Purchases(loginId, ingredientId, purchaseDate, price, quantity)

SQL DDL Code

```
-- Project Phase 2
-- By Jack Berry, Jonathon Houston, and Matt Settle-Rakay
-- Due June 8th @ 11:59PM EST
-- Purpose: converting the ER schema into a relational

-- Debugging Lines (character casing in line with table creation)
DROP TABLE Purchases CASCADE CONSTRAINTS;
DROP TABLE Requires CASCADE CONSTRAINTS;
DROP TABLE Invents CASCADE CONSTRAINTS;
DROP TABLE CreatedUsing CASCADE CONSTRAINTS;
DROP TABLE Creates CASCADE CONSTRAINTS;
DROP TABLE SoldTo CASCADE CONSTRAINTS;
DROP TABLE ProductBatch CASCADE CONSTRAINTS;
DROP TABLE Ingredient CASCADE CONSTRAINTS;
DROP TABLE Recipe CASCADE CONSTRAINTS;
DROP TABLE Worker CASCADE CONSTRAINTS;
DROP TABLE Customer CASCADE CONSTRAINTS;

-- Table Creation
CREATE TABLE Customer (
    customerId CHAR(8) PRIMARY KEY,
    cardNo CHAR(10),
    purchaseHistory VARCHAR(500),
    rewardsId CHAR(8),
    points NUMBER(6),
    membershipDate DATE
);

CREATE TABLE Worker (
    loginId CHAR(8) PRIMARY KEY,
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    taxId CHAR(9),
    availability VARCHAR(50),
    QualificationList VARCHAR(500),
    title VARCHAR(50),
    name VARCHAR(50)
);

CREATE TABLE ProductBatch (
    batchId CHAR(8) PRIMARY KEY,
    quantity NUMBER(6),
    bestBy DATE,
    pricePerUnit NUMBER(6,2)
);

CREATE TABLE Recipe (
    recipeId CHAR(8) PRIMARY KEY,
    temperature NUMBER(4),
    Allergen VARCHAR(500),
    time NUMBER(4),
    Instructions VARCHAR(500),
    Name VARCHAR(50)
);

CREATE TABLE Ingredient (
    ingredientId CHAR(8) PRIMARY KEY,
    quantity NUMBER(8,2),
    bestBefore DATE,
    name VARCHAR(50),
    Supplier VARCHAR(50),
    SupplierContact VARCHAR(50)
);

CREATE TABLE SoldTo (
    customerId CHAR(8),
    batchId CHAR(8),
    dateSold DATE,
    price NUMBER(6,2),
    method VARCHAR(20),
    PRIMARY KEY (customerId, batchId, dateSold),
    FOREIGN KEY (customerId) REFERENCES Customer(customerId),
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```
    FOREIGN KEY (batchId) REFERENCES ProductBatch(batchId)
);

CREATE TABLE Creates (
    loginId CHAR(8),
    batchId CHAR(8),
    PRIMARY KEY (loginId, batchId),
    FOREIGN KEY (loginId) REFERENCES Worker(loginId),
    FOREIGN KEY (batchId) REFERENCES ProductBatch(batchId)
);

CREATE TABLE CreatedUsing (
    batchId CHAR(8),
    recipeId CHAR(8),
    DateCreated DATE,
    Quantity NUMBER(6),
    PRIMARY KEY (batchId, recipeId),
    FOREIGN KEY (batchId) REFERENCES ProductBatch(batchId),
    FOREIGN KEY (recipeId) REFERENCES Recipe(recipeId)
);

CREATE TABLE Invents (
    loginId CHAR(8),
    recipeId CHAR(8),
    DateInvented DATE,
    PRIMARY KEY (loginId, recipeId),
    FOREIGN KEY (loginId) REFERENCES Worker(loginId),
    FOREIGN KEY (recipeId) REFERENCES Recipe(recipeId)
);

CREATE TABLE Requires (
    recipeId CHAR(8),
    ingredientId CHAR(8),
    amount VARCHAR(30),
    PRIMARY KEY (recipeId, ingredientId),
    FOREIGN KEY (recipeId) REFERENCES Recipe(recipeId),
    FOREIGN KEY (ingredientId) REFERENCES Ingredient(ingredientId)
);

CREATE TABLE Purchases (
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    loginId CHAR(8),
    ingredientId CHAR(8),
    purchaseDate DATE,
    price NUMBER(6,2),
    quantity NUMBER(8,2),
    PRIMARY KEY (ingredientId, purchaseDate),
    FOREIGN KEY (loginId) REFERENCES Worker(loginId),
    FOREIGN KEY (ingredientId) REFERENCES Ingredient(ingredientId)
);

-- Test Data Creation
INSERT INTO Customer VALUES ('CUST0001', 'CARD100001', 'Cookies, Bread',
'RWDS0001', 120, DATE '2024-01-15');
INSERT INTO Customer VALUES ('CUST0002', 'CARD100002', 'Cupcakes',
'RWDS0002', 45, DATE '2024-01-15');
INSERT INTO Customer VALUES ('CUST0003', 'CARD100003', 'Bread', null,
null, DATE '2024-01-15');


INSERT INTO Worker VALUES (
    'WORK0001',
    '111223333',
    'Morning',
    'Bread Baking, Food Safety, Pastry Preparation',
    'Baker',
    'Ava Miller'
);

INSERT INTO Worker VALUES (
    'WORK0002',
    '222334444',
    'Afternoon',
    'Inventory Management, Ordering, Food Safety',
    'Manager',
    'Liam Carter'
);

INSERT INTO Worker VALUES (
    'WORK0003',
    '333445555',
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```
    'Evening',
    'Cake Decorating, Dough Preparation',
    'Assistant',
    'Noah Brooks'
);

INSERT INTO ProductBatch VALUES ('BACH0001', 36, DATE '2026-06-15', 2.50);
INSERT INTO ProductBatch VALUES ('BACH0002', 20, DATE '2026-06-18', 6.00);
INSERT INTO ProductBatch VALUES ('BACH0003', 24, DATE '2026-06-16', 3.25);

INSERT INTO Recipe VALUES (
    'RECP0001',
    350,
    'Gluten, Dairy',
    12,
    'Mix dough, add chocolate chips, portion onto tray, and bake.',
    'Chocolate Chip Cookies'
);

INSERT INTO Recipe VALUES (
    'RECP0002',
    425,
    'Gluten',
    35,
    'Prepare starter, mix dough, proof, shape, and bake.',
    'Sourdough Bread'
);

INSERT INTO Recipe VALUES (
    'RECP0003',
    350,
    'Eggs, Dairy, Gluten',
    18,
    'Mix batter, pour into cups, bake, and cool before frosting.',
    'Vanilla Cupcakes'
);

INSERT INTO Ingredient VALUES (
    'INGR0001',
    100.00,
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```
    DATE '2026-12-01',
    'Flour',
    'Bakery Supply Co',
    'supply@example.com'
);

INSERT INTO Ingredient VALUES (
    'INGR0002',
    80.00,
    DATE '2027-01-15',
    'Sugar',
    'Sweet Wholesale',
    'sweet@example.com'
);

INSERT INTO Ingredient VALUES (
    'INGR0003',
    40.00,
    DATE '2026-09-10',
    'Chocolate Chips',
    'Cocoa Direct',
    'cocoa@example.com'
);

INSERT INTO Ingredient VALUES (
    'INGR0004',
    120.00,
    DATE '2026-06-20',
    'Eggs',
    'Farm Fresh',
    'farm@example.com'
);

INSERT INTO Ingredient VALUES (
    'INGR0005',
    50.00,
    DATE '2026-07-05',
    'Butter',
    'Dairy Best',
    'dairy@example.com'
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);

INSERT INTO SoldTo VALUES ('CUST0001', 'BACH0001', DATE '2026-06-04',
7.50, 'Credit');
INSERT INTO SoldTo VALUES ('CUST0002', 'BACH0002', DATE '2026-06-04',
12.00, 'Cash');
INSERT INTO SoldTo VALUES ('CUST0003', 'BACH0003', DATE '2026-06-05',
6.50, 'Debit');

INSERT INTO Creates VALUES ('WORK0001', 'BACH0001');
INSERT INTO Creates VALUES ('WORK0001', 'BACH0002');
INSERT INTO Creates VALUES ('WORK0003', 'BACH0003');

INSERT INTO CreatedUsing VALUES ('BACH0001', 'RECP0001', DATE
'2026-06-04', 36);
INSERT INTO CreatedUsing VALUES ('BACH0002', 'RECP0002', DATE
'2026-06-04', 20);
INSERT INTO CreatedUsing VALUES ('BACH0003', 'RECP0003', DATE
'2026-06-05', 24);

INSERT INTO Invents VALUES ('WORK0001', 'RECP0001', DATE '2025-10-15');
INSERT INTO Invents VALUES ('WORK0002', 'RECP0002', DATE '2025-11-02');
INSERT INTO Invents VALUES ('WORK0003', 'RECP0003', DATE '2026-01-10');

INSERT INTO Requires VALUES ('RECP0001', 'INGR0001', '2 cups');
INSERT INTO Requires VALUES ('RECP0001', 'INGR0002', '1 cup');
INSERT INTO Requires VALUES ('RECP0001', 'INGR0003', '1 cup');
INSERT INTO Requires VALUES ('RECP0001', 'INGR0005', '1 stick');

INSERT INTO Requires VALUES ('RECP0002', 'INGR0001', '4 cups');

INSERT INTO Requires VALUES ('RECP0003', 'INGR0001', '2 cups');
INSERT INTO Requires VALUES ('RECP0003', 'INGR0002', '1.5 cups');
INSERT INTO Requires VALUES ('RECP0003', 'INGR0004', '3 eggs');
INSERT INTO Requires VALUES ('RECP0003', 'INGR0005', '1 stick');

INSERT INTO Purchases VALUES ('WORK0002', 'INGR0001', DATE '2026-06-01',
30.00, 50.00);
INSERT INTO Purchases VALUES ('WORK0002', 'INGR0002', DATE '2026-06-01',
24.00, 40.00);
```

```sql
INSERT INTO Purchases VALUES ('WORK0002', 'INGR0003', DATE '2026-06-02',
45.00, 20.00);
INSERT INTO Purchases VALUES ('WORK0002', 'INGR0004', DATE '2026-06-02',
18.00, 60.00);
INSERT INTO Purchases VALUES ('WORK0002', 'INGR0005', DATE '2026-06-03',
35.00, 25.00);

-- Test Queries (generated with ai but not needed for this portion of the
assignment)
SELECT * FROM Customer;
SELECT * FROM RewardsMember;
SELECT * FROM Joins;
SELECT * FROM Worker;
SELECT * FROM ProductBatch;
SELECT * FROM Recipe;
SELECT * FROM Ingredient;
SELECT * FROM SoldTo;
SELECT * FROM Creates;
SELECT * FROM CreatedUsing;
SELECT * FROM Invents;
SELECT * FROM Requires;
SELECT * FROM Purchases;

SELECT
    Customer.customerId,
    Customer.cardNo,
    ProductBatch.batchId,
    SoldTo.dateSold,
    SoldTo.price,
    SoldTo.method
FROM Customer
JOIN SoldTo ON Customer.customerId = SoldTo.customerId
JOIN ProductBatch ON SoldTo.batchId = ProductBatch.batchId;

SELECT
    Recipe.Name AS recipeName,
    Ingredient.name AS ingredientName,
    Requires.amount
FROM Recipe
JOIN Requires ON Recipe.recipeId = Requires.recipeId
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```
JOIN Ingredient ON Requires.ingredientId = Ingredient.ingredientId
ORDER BY Recipe.Name, Ingredient.name;

SELECT
    Worker.name AS workerName,
    Recipe.Name AS recipeName,
    Invents.DateInvented
FROM Worker
JOIN Invents ON Worker.loginId = Invents.loginId
JOIN Recipe ON Invents.recipeId = Recipe.recipeId;
```